ESCUELA COLOMBIANA DE INGENIERÍA JULIO GARAVITO
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Linux Deep Dive

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What is Linux?

- Open source Operative System (OS)
- OS is a resource manager
  - Processor time
  - Memory
  - Storage space
- Kernel
  - OS core
  - OS's lowest level

What is a Linux distro?

- Desktop environment
- Package manager
- Apps
- Objectives and philosophy

- *Think of Linux Distros as ice cream flavors.*

CentOS

ubuntu

Why use Linux in cybersecurity?

- Linux is the most used operating system by cybersecurity professionals due to its control, flexibility, and its great tool ecosystem.
- Kali Linux is a specialized distro based on Debian that includes tolos for pentesting, forensics, network analysis, reverse engineering, etc.
- Working with the terminal (CLI) directly allows for task automation, direct interaction with tools and a better understanding of how systems work.

Linux Filesystem Structure

Directory	Typical contents
/	System root
/home	Users' personal directories
/etc	System configuration files
/bin	Essential system commands
/usr	User applications
/var	Variable data (logs, cache)
/tmp	Temporary files
/root	root user's home directory
/dev	Devices (disks, USB, etc.)
/proc	Information about processes and the kernel

Essential navigation and manipulation commands

Navigation and exploration

```
pwd          # Show the current directory
ls           # List files
cd           # Change directory
```

File and directory manipulation

```
cp file destination           # Copy
mv file destination           # Move or rename
rm file                       # Delete
mkdir directory_name          # Create directory
touch file.txt                # Create an empty file
cat file.txt                  # See contents of a file
less file.txt                 # See paginated contents of a file
```

Useful exploration

```
tree                          # See directory structure
file file                     # Detect file type
find /path -name "*.txt" # Find files (in this case, by extension/name)
```

File permissions and user management

Understanding permissions

```
ls -l

# Typical output
-rwxr-xr-- 1 user group 1234 jul 18 file.sh
```

The first 10 characters indicate type and permissions:

- `r` = read
- `w` = write
- `x` = execute

Structure: `[user][group][others]`

Permissions and user commands

```
chmod 755 file.sh            # Change permissions (rwxr-xr-x)
chown user:group file        # Change owner
whoami                       # Show current user
sudo command                 # Execute with privileged permissions
```

Processes and services

Process monitoring

```
ps aux              # See all processes
top / htop          # Dynamic view (htop must be installed)
kill PID            # End process
```

Services (systemd)

```
systemctl status service_name
systemctl start/stop/restart service_name
```

Package installation and use of man

Package installation

```
sudo apt update                # Update the packages list
sudo apt install name          # Install a package
sudo apt remove name           # Remove a package
```

Get help

```
man command       # Complete manual
command --help    # Quick help
```

Basic concepts of Bash scripting

Basic concepts of Bash scripting

Scripts can automate tasks using command tools.

```bash
#!/bin/bash
echo "Hello, World"
```

Useful script commands

- `if, then, else`
- `for, while`
- **`read`** (read user input)
- **`$(command)`** (command substitution)

Simple example

```bash
#!/bin/bash
echo "What is your name?"
read name
echo "Hello, $name"
```

Allow execution

```
chmod +x script.sh
./script.sh
```

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Gracias